Electronic Supplementary Information

Geometric algebra meets Cartesian tensors: higher-order equivariance for interatomic potentials

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S1 Background on Cl(3,0) and the geometric product

The real Clifford algebra Cl(3,0) is the eight-dimensional associative unital algebra generated by an orthonormal basis $\{e_1,e_2,e_3\}$ of $\mathbb{R}^3$ subject to $e_ie_j+e_je_i=2\delta_{ij}$ [1]. A basis of Cl(3,0) ordered by grade is $\{1,\ e_1,e_2,e_3,\ e_{12},e_{13},e_{23},\ e_{123}\}$, corresponding respectively to a scalar (grade 0), three vectors (grade 1), three bivectors (grade 2), and a pseudoscalar (grade 3), so any multivector $X\in\mathrm{Cl}(3,0)$ has a unique grade decomposition $X=X^{(0)}+X^{(1)}+X^{(2)}+X^{(3)}$ [2].

The geometric product is the bilinear, associative, unital extension of the generator relations, with the vector decomposition [3] below: for two vectors $u,v\in\mathbb{R}^3$ embedded in Cl(3,0) it decomposes as $uv=u\cdot v+u\wedge v$, where $u\cdot v=\frac{1}{2}(uv+vu)=\sum_i u_iv_i$ is the symmetric scalar part and $u\wedge v=\frac{1}{2}(uv-vu)=\sum_{i<j}(u_iv_j-u_jv_i)e_{ij}$ is the antisymmetric bivector part.

With $I=e_1e_2e_3$, $I^2=-1$ and $I^{-1}=-I$, we adopt the bivector dual convention $\star\alpha=\alpha I^{-1}$ [4]. Direct computation yields the sign convention

$$\star e_{12}=+e_3,\qquad \star e_{13}=-e_2,\qquad \star e_{23}=+e_1,$$

which gives $\star(u\wedge v)=u\times v$ by substitution. The dual is rotationally equivariant, but its vector output is axial: under inversion, $e_ie_j\to(-e_i)(-e_j)=e_ie_j$, giving parity $+1$ rather than the polar-vector parity -1 [5]. We apply this distinction to the readout in Section S4.

The geometric product is SO(3)-equivariant under the rotor action: for any rotor $R=\exp(-\theta B/2)$ built from a unit bivector B and any multivectors a,b, conjugation $a\mapsto RaR^{-1}$ preserves the product, $R(ab)R^{-1}=(RaR^{-1})(RbR^{-1})$ [6]. This product-preserving action underlies the Clifford track.

S2 Architectural comparison with ICTP

ICTP builds an O(3)-equivariant architecture from irreducible Cartesian tensors, combines ranks through irreducible tensor products, and constructs higher-body-order features through repeated products [7]. CliffordSTF shares the STF foundation but retains a parallel Clifford substrate; the following four points specify our design choices.

Algebraic substrate. CliffordSTF runs a Clifford Cl(3,0) multivector track in parallel with the STF tracks, and the Clifford track carries both scalars and pseudoscalars that have a natural parity structure through the grade involution, as well as bivectors that expose an extra L=1 axial channel via the Hodge star. The STF tracks in CliffordSTF supply the L=2 and L=3 components that the Clifford geometric product cannot reach, not the entire equivariant substrate.

Bilinear operation. CliffordSTF uses two different bilinears in tandem: the Clifford geometric product acting on the 8-dimensional multivector via an 8×8 Cayley table, and a small set of closed-form symmetric-traceless contractions acting on the STF tracks. The Clifford product handles scalar, vector, bivector, and pseudoscalar channels with a single contraction; the STF operations handle rank-2 and rank-3 tensors through explicit symmetrisation and trace removal. The two are coupled bijectively to Clebsch–Gordan channels as specified in Theorem 1 and its higher-order extensions.

Cross-track coupling. In CliffordSTF the Clifford and STF tracks are coupled bidirectionally: the STF_2 track is generated from grade-1 vector features via $\mathrm{STF}_2(\hat{r}_{ij},h_j^{\mathrm{vec}})$, and is fed back into the grade-1 vector component of the Clifford message through the $S\cdot v$ contraction. The `stf2_no_cross` variant does not regress relative to the CliffordIP base: it reaches f_{cos}=0.456 and e_{MAE}=25.8 meV against 0.030 and 125.8 meV for vanilla L=1 Clifford (Table S2), because the L=2 edge descriptor supplies angular content to message passing whether or not the tracks are coupled. The full $\mathrm{STF}_2+\mathrm{STF}_3$ configuration has one fewer divergent run with coupling and a lower median energy MAE in this sample; paired statistics are reported in Section S18.

Equivariance scope. CliffordSTF proves SO(3) equivariance; under O(3), the polar-vector grade-1 component and the axial-vector Hodge-of-bivector component would need distinct parity labels 1^- and 1^+ respectively (Section 4.3, Section S4). The current implementation concatenates the two in the force readout under the SO(3) framework, which is consistent under rotations but would need refactoring for full O(3).

CliffordSTF retains geometric-product message passing while adding STF features through explicit cross-track maps. The ablation in Section 5.3 evaluates the contribution of each component.

S3 Rotational content of projective multivectors

GATr [8] uses the 16-dimensional projective algebra Cl(3,0,1). Applying the rotation action to its basis, the null basis vector is unchanged; the decomposition of main-text eqn (3) therefore occurs twice, giving $4\cdot(L{=}0)\oplus4\cdot(L{=}1)$. Thus neither multivector representation contains an explicit $L\geq2$ irrep. This statement concerns the representation carried by the multivector, rather than the expressivity of a complete nonlinear network.

S4 SO(3) versus O(3) equivariance

All guarantees stated in §4 are SO(3)-equivariance guarantees. Polar and axial vectors carry equivalent L=1 representations under SO(3), but differ by a factor of $\det(R)$ under $R \in \mathrm{O}(3)$, following the parity convention used in `e3nn`[5]. Consequently, our rotational readout can concatenate them, while reflection-equivariant mixing would require the distinct labels in Table S1.

Table S1 O(3) parity labels for the tracks used in CliffordSTF. The general rule is STF_ℓ built from polar vectors has parity $(-1)^\ell$, matching the parity of the spherical harmonic $Y_{\ell m}(-\hat{n}) = (-1)^\ell Y_{\ell m}(\hat{n})$[9].

Object	Physics analog	Parity	O(3) label
Grade-0 scalar	charge q	$+$	0^+
Grade-1 vector e_i	position $\mathbf{r}$	$-$	1^- (polar)
Grade-2 bivector e_{ij} / Hodge dual	angular momentum $\mathbf{L}=\mathbf{r}\times\mathbf{p}$	$+$	1^+ (axial)
Grade-3 pseudoscalar I	triple product $\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$	$-$	0^-
STF_2 from polar vectors	quadrupole Q_{ij}	$+$	2^+
STF_3 from polar vectors	octupole O_{ijk}	$-$	3^-

Extending our model to full O(3) equivariance would require distinguishing the 1^-, 1^+, 2^+ and 3^- tracks and using parity-compatible mixing and readout, following the parity-aware tensor-product construction used in ICTP[7].

S5 Derivation of the STF coefficients

We work through the classical STF projection[10] to recover the three rational coefficients used in §4.1.

STF_2 detracing coefficient $\frac{1}{3}$. In d dimensions, the symmetric traceless part of a rank-2 tensor requires subtracting $\frac{1}{d}\delta_{ij}T_{kk}$ to enforce $\sum_i T^{\text{sym-traceless}}_{ii} = 0$[11]. Direct check in $d=3$: $\delta_{ii}\mathrm{STF}_2(u,v)_{ii} = (u\cdot v) - \frac{1}{3}(3)(u\cdot v) = 0$.

STF_3 trace vector coefficient $\frac{2}{3}$. Since S_{ij} is symmetric traceless ($S_{ii}=0$), the trace of T^{sym}_{ijk} over any pair of indices reduces to two of the three summands:

$$T^{\text{sym}}_{iik} = \tfrac{1}{3}(S_{ii}v_k + S_{ik}v_i + S_{ik}v_i) = \tfrac{1}{3}(0 + 2S_{kl}v_l) = \tfrac{2}{3}S_{kl}v_l.$$

By full symmetry, $T^{\text{sym}}_{iki} = T^{\text{sym}}_{kii} = A_k$ as well.

STF_3 detracing coefficient $\frac{1}{5}$. Set $U_{ijk} = T^{\text{sym}}_{ijk} - \alpha(\delta_{ij}A_k + \delta_{ik}A_j + \delta_{jk}A_i)$ and require $U_{iik} = 0$. Computing the trace:

$$U_{iik} = A_k - \alpha(3A_k + A_k + A_k) = A_k - 5\alpha A_k \implies \alpha = \tfrac{1}{5}.$$

More generally, the three-dimensional per-delta first-trace coefficient at rank ℓ is $1/(2\ell-1)$, yielding $\frac{1}{3}$ at ℓ=2, $\frac{1}{5}$ at ℓ=3, $\frac{1}{7}$ at ℓ=4[12]. Applequist gives the fully detraced closed form for arbitrary rank[13].

Degrees-of-freedom check for STF_3. The symmetric rank-3 tensor T^{sym} in $\mathbb{R}^3$ has $\binom{3+3-1}{3} = 10$ independent components; subtracting the 3-component trace vector A leaves $10-3=7=2\ell+1$ for $\ell=3$, matching the dimension of the L=3 irrep of SO(3)[14].

S6 Explicit coordinate formulas for the constructive CG maps

We enumerate every constructive map used in the architecture and verified in Section S7. The formulas use classical irreducible Cartesian tensor coupling[10] and the STF coefficients derived in Section S5.

$\mathbf{1}\otimes\mathbf{1}\to\mathbf{0}\oplus\mathbf{1}\oplus\mathbf{2}$. For $u,v\in\mathbb{R}^3$:

- L=0: $(u\otimes v)_{L=0} = u\cdot v = u_i v_i$.
- L=1: $(u\otimes v)_{L=1} = \star(u\wedge v)_i = \varepsilon_{ijk}u_j v_k$, equivalently the cross product $u\times v$.
- L=2: $(u\otimes v)_{L=2} = \mathrm{STF}_2(u,v)_{ij} = \frac{1}{2}(u_i v_j + u_j v_i) - \frac{1}{3}(u\cdot v)\delta_{ij}$.

$\mathbf{2}\otimes\mathbf{1}\to\mathbf{1}\oplus\mathbf{2}\oplus\mathbf{3}$. For a symmetric traceless $S\in\mathbb{R}^{3\times3}$ and $v\in\mathbb{R}^3$:

- L=1: $(S\cdot v)_i = S_{ij}v_j$.
- L=2: symmetric traceless part of $\varepsilon_{ikl}S_{jl}v_k$, i.e. $[\mathrm{sym}(S\times v) - \frac{1}{3}\mathrm{tr}(\cdot)\mathbb{I}]$; explicit 5-component formulas are in the code release (`contract_stf2_vec_to_stf2`).
- L=3: $\mathrm{STF}_3(S,v)_{ijk} = S_{(ij}v_{k)} - \frac{1}{5}(\delta_{ij}A_k + \delta_{ik}A_j + \delta_{jk}A_i)$ with $A_k = \frac{2}{3}S_{kl}v_l$; the coefficients $\frac{2}{3}$ and $\frac{1}{5}$ follow from the trace calculations in Section S5.

$\mathbf{2}\otimes\mathbf{2}\to\mathbf{0}\oplus\mathbf{1}\oplus\mathbf{2}\oplus\mathbf{3}$ (with L=4 omitted). For symmetric traceless S_1, S_2:

- L=0: $\langle S_1, S_2\rangle_F = S_{1,ij}S_{2,ij}$ (Frobenius inner product).

- L=1: antisymmetric contraction $\varepsilon_{ijk} S_{1,jl} S_{2,kl}$.
- L=2: symmetric traceless projection of $S_{1,ik} S_{2,kj}$.
- L=3: an analogous symmetric-then-detraced construction treating one operand as a vector-indexed rank-2 tensor.
- L=4: outside our $L_{\max}$=3 cut and not used.

$\mathbf{3} \otimes \mathbf{1} \to \mathbf{2} \oplus \mathbf{3}$ (with L=4 omitted). For a symmetric traceless rank-3 tensor T and vector v:

- L=2: $(T \cdot v)_{ij} = T_{ijk} v_k$ followed by trace removal (`contract_stf3_vec_to_stf2`).
- L=3: analogous symmetric-then-detraced construction.
- L=4: omitted.

Each map is a polynomial in its inputs with constant rational coefficients. There are no learned Clebsch–Gordan coefficients, no Wigner-D tables, and no calls to `e3nn`[5]. All tensor products of SO(3) irreps above are multiplicity-free, consistent with the general SU(2) fact that $L_1 \otimes L_2 = \bigoplus_{L=|L_1-L_2|}^{L_1+L_2} L$ each with multiplicity one[9].

S7 Numerical verification of the constructive CG maps and end-to-end equivariance

Primitive-level verification. The released tests compare the rank-2 and rank-3 STF products with explicit Cartesian symmetrisation and trace removal on 10^3 random double-precision inputs. They also check that these products commute with proper rotations, using an absolute tolerance of 10^{-11}. Separate checks cover geometric-product rotation covariance, grade-norm invariance and reverse involution. These checks test the implemented Cartesian identities; they do not constitute a comparison of every admissible coupling with Wigner $3j$-symbols.

End-to-end checks. The released tests apply a proper rotation and translation to a non-collinear molecular geometry, rebuild its radius graph, and compare energies and rotated forces at absolute tolerance 10^{-3} for the twelve ablation settings. They also test atom-permutation symmetry. For the energy-gradient model, finite-difference checks compare forces with $-\partial E/\partial x$ and test zero net force and torque. The tests are available in `tests/test_physics.py` and `tests/test_models_algebra.py`.

S8 Edge embedding

We expand pairwise distances $d_{ij} = \|\mathbf{r}_{ij}\|$ through Gaussian radial basis functions (RBFs), as in SchNet[15], with centres evenly spaced between 0 and our cutoff r_c, and apply the cosine envelope $\phi_c(d_{ij}) = \frac{1}{2}[\cos(\pi d_{ij}/r_c) + 1]$ that drives edge features smoothly to zero at the cutoff. Two small MLPs map the RBF expansion to channel-wise scalar weights for the grade-0 part and channel-wise weights for the grade-1 part of the edge multivector; the grade-1 direction $\hat{r}_{ij}$ is then multiplied by the channel weights and placed in the vector component. For CliffordSTF variants with stf_mode $\neq$ none, a third MLP produces channel weights for the STF_2 edge feature, populated with $\mathrm{STF}_2(\hat{r}_{ij}, \hat{r}_{ij})$ (a canonical L=2 descriptor of the edge direction). When stf_mode = stf2+stf3, the STF_3 edge block is initialised to zero and is populated only via message passing within node features.

S9 Augmented product at the many-body stage

At the many-body stage of each interaction layer, an augmented product couples the Clifford and STF tracks through six bilinear terms. Given operands a and b each carrying Clifford, STF_2, and (optionally) STF_3 components, the augmented product computes:

- The ordinary geometric product $a^{\mathrm{Cl}} b^{\mathrm{Cl}}$ via the 8×8 Cayley table;
- A new STF_2 from the grade-1 components via $\mathrm{STF}_2(a^{\mathrm{vec}}, b^{\mathrm{vec}})$;
- An updated STF_2 via the L=2 $\otimes$ L=1 $\to$ L=2 path, applied from either operand (symmetrised-then-detraced cross-contraction of a^{STF_2} with b^{vec}, and vice versa);
- An L=0 contribution to grade-0 via $\langle a^{\mathrm{STF}_2}, b^{\mathrm{STF}_2} \rangle_F = a^{\mathrm{STF}_2}_{ij} b^{\mathrm{STF}_2}_{ij}$;
- A new STF_3 via $\mathrm{STF}_3(a^{\mathrm{STF}_2}, b^{\mathrm{vec}})$;
- An STF_2 update via $\mathrm{STF}_3 \otimes L$=1 $\to L$=2 contraction of a^{STF_3} against b^{vec}.

Each term corresponds to a CG channel in one of the decompositions $\mathbf{1} \otimes \mathbf{1}$, $\mathbf{2} \otimes \mathbf{1}$, $\mathbf{2} \otimes \mathbf{2}$, or $\mathbf{3} \otimes \mathbf{1}$ enumerated in Section S6. These bilinears provide gradient paths between the Clifford and STF tracks. Their empirical effects are assessed in Section 5.3 and Section S18.

S10 Equivariant normalisation and gating

Each CliffordSTFLinear layer is block-diagonal across tracks: it applies independent linear maps of shape $C_{\text{in}} \to C_{\text{out}}$ over channels within the Clifford multivector, the STF_2 tensor, and the STF_3 tensor, without mixing them linearly. Cross-track information flow occurs only through the bilinear products described in §4.2.

Track-wise equivariant normalisation rescales each track by a per-channel invariant norm. For the Clifford track we compute the squared per-grade Clifford norm $\|h^{(k)}\|^2$ as the scalar part of $h^{(k)}\widetilde{h^{(k)}}$, where $\widetilde{\cdot}$ is the reverse[4]; for STF_2 we use the Frobenius norm $\|S\|_F^2 = S_{ij}S_{ij}$ with $S_{zz} = -S_{xx} - S_{yy}$ substituted; for STF_3 we use the analogous Frobenius norm with multiplicity factors reflecting the symmetric tensor structure. These scalar contractions are rotationally invariant, as required for equivariant norm-based nonlinearities[6,7].

Our activation applies SiLU[16] to grade-0 features and a sigmoid-gated rescaling to each $L \geq 1$ track: for a feature f at rank $\ell \geq 1$, we compute $g = \sigma(\text{MLP}(\|f\|))$ with the sigmoid MLP acting on the invariant norm, and output $g \cdot f$. Multiplication by an invariant scalar gate preserves equivariance, following the gating constructions of Weiler *et al.* [17], Schütt *et al.* [18]; we apply this principle separately to the three tracks.

S11 Adaptive L-track routing

Our adaptive routing modulates the STF tracks using invariant gates. It is applied as a per-atom, per-track multiplicative gate $g_\ell \in [0,1]$ between message-passing layers, rescaling the STF_ℓ track as $h_i^{\text{STF}_\ell} \leftarrow g_\ell^{(i)} \cdot h_i^{\text{STF}_\ell}$. The Clifford track is always active. Three modes are supported.

- **None.** All atoms use all tracks; $g_\ell^{(i)} = 1$.
- **Static.** Gates are learned per atomic type via a small embedding table.
- **Learned.** Gates are produced by a two-layer MLP with sigmoid output applied to four per-atom invariant descriptors: (i) coordination number $|\mathcal{N}(i)|$, (ii) angular variance $1 - \|\bar{r}_i\|$ where $\bar{r}_i = \frac{1}{|\mathcal{N}(i)|}\sum_{j\in\mathcal{N}(i)} \hat{r}_{ij}$, (iii) mean neighbour distance, (iv) grade-0 feature norm averaged over channels. All four descriptors are $\text{SO}(3)$-invariant, so the gate is invariant and the gated product remains equivariant.

S12 Full ablation configurations

Table S2 lists eleven variants plus the CliffordIP baseline. The configurable components are STF node ranks (none, rank 2, or ranks 2 and 3), Hodge readout, cross-track coupling, and routing (off, static, or learned). The rank-2 edge descriptor is controlled separately by `use_l2`: it is enabled in every CliffordSTF variant, including those without STF node tracks, and disabled in the CliffordIP baseline. Thus the edge-only variant isolates the descriptor, while the "Hodge only" row adds Hodge readout to that same configuration.

The STF-only readout variant retains STF tracks and cross-track coupling in message passing but removes Clifford contributions from both output heads. Energy is read from STF invariants and forces from the per-edge $S\hat{r}$ contraction. The configuration columns in Table S2 retain every flag setting; Section 5.3 reports the key comparisons and Section S18 gives the per-seed coupling analysis.

Table S2 Full rMD17 ablation (eleven variants plus CliffordIP; mean ± standard deviation over six seeds on aspirin, benzene, ethanol and salicylic acid; best value per column in bold, second best underlined). Edge denotes the rank-2 edge descriptor; tracks denote STF node features; Hodge denotes the axial-vector readout; Cross denotes cross-track coupling. Routing and Mode specify the invariant gates; Clifford readout indicates whether Clifford features enter the force head.

	Configuration							Metrics		
Variant	L=2 edge	STF tracks	Hodge	Cross	Routing	Mode	Clifford readout	f_{MAE} (meV Å^{-1})	e_{MAE} (meV)	$f_{\cos}\uparrow$
Full	on	stf2+stf3	on	on	on	learned	on	16.802 ± 0.556	45.986 ± 48.658	**0.506** ± 0.045
STF2 static routing	on	stf2	on	on	on	static	on	16.836 ± 0.640	26.482 ± 6.624	<u>0.504</u> ± 0.041
L=2 edge descriptor only	on	none	off	off	off	—	on	17.221 ± 0.537	<u>26.294</u> ± 8.464	0.498 ± 0.084
STF2 learned routing	on	stf2	on	on	on	learned	on	<u>16.768</u> ± 1.202	37.778 ± 13.440	0.492 ± 0.100
STF2+STF3	on	stf2+stf3	on	on	off	—	on	17.029 ± 1.016	36.792 ± 9.466	0.489 ± 0.119
Full no cross	on	stf2+stf3	on	off	off	—	on	17.170 ± 0.674	70.409 ± 92.001	0.474 ± 0.072
Hodge only	on	none	on	off	off	—	on	**16.584** ± 1.166	30.975 ± 13.434	0.472 ± 0.103
STF2 no hodge	on	stf2	off	on	off	—	on	17.688 ± 0.251	36.261 ± 25.029	0.460 ± 0.067
STF2 no cross	on	stf2	on	off	off	—	on	17.317 ± 0.863	**25.790** ± 13.248	0.456 ± 0.095
STF2	on	stf2	on	on	off	—	on	17.511 ± 0.947	34.909 ± 16.325	0.421 ± 0.107
STF-only readout	on	stf2+stf3	off	on	off	—	off	19.545 ± 0.098	36.640 ± 39.015	0.097 ± 0.028
CliffordIP ($L\leq1$)	off	—	—	—	—	—	on	19.704 ± 0.043	125.848 ± 77.191	0.030 ± 0.003

S13 Training protocol

CliffordSTF is trained under the protocol of the CliffordIP study[19], whose published values supply every baseline in Tables 1 and 3: Adam[20] at a constant learning rate of 10^{-4}, no weight decay, no weight averaging or warm-up, early stopping with patience 30, a cutoff

of 6.0 Å with at most 50 neighbours per atom, and the same data splits (rMD17 900/100/1,000 random split; OC20 200K S2EF and IS2RE training sets; full OC22 training sets). The rMD17 and OC20/OC22 CliffordSTF models have 1.06×10^6 parameters, within the $\sim 10^6$-parameter regime specified above. Two settings differ from the CliffordIP runs: CliffordSTF clips the gradient norm at 10, which the CliffordIP runs did not, and its OC22 IS2RE budget is 200 epochs with patience 30, against 50 epochs in the CliffordIP runs[19]. Table S3 lists the per-benchmark settings. All CliffordSTF values are averaged over every seed that was run.

Table S3 Per-benchmark training settings for CliffordSTF. λ_F is the force-loss weight. All runs use early-stopping patience 30 and gradient-norm clipping at 10.

Benchmark	Task	Epochs	Batch	Loss	λ_F	Seeds
rMD17	energy + forces (10 molecules)	250	32	MSE	1.0	4
OC20	S2EF	50	16	per-atom MAE (E) + ℓ_2 (F)	3.0	3
OC20	IS2RE	50	16	L1	—	3
OC22	S2EF	50	16	per-atom MAE (E) + ℓ_2 (F)	3.0	4
OC22	IS2RE	200	16	L1	—	4

Table S4 CliffordSTF architecture hyperparameters. Cutoffs in Å; force formulation and training settings are specified in the text.

Configuration	Benchmarks	Architecture
CliffordSTF (1.06×10^6)	rMD17, OC20, OC22	Cutoff 6; channels 60; layers 5; RBFs 50; STF ranks 2 and 3; learned adaptive routing, cross coupling and Hodge force head on; GP readout and self-interaction off.
CliffordSTF-1.9M	rMD17 (Section S15)	Cutoff 5; channels 80; output MLP width 96; other architecture settings as above; energy-gradient forces.
CliffordSTF-10M	rMD17 (Section S15)	Cutoff 6; channels 188; layers 5; output MLP width 256; other architecture settings as above; energy-gradient forces.

S14 OC20 and OC22 full results

Tables S5–S6 report every validation split. Baseline values are the published values of the CliffordIP study[19]; CliffordSTF is averaged over all of its seeds.

S15 rMD17 capacity scaling and comparison with published results

Configurations. The capacity-scaling result of Section 5.5 uses two larger CliffordSTF configurations, trained with recipes of the kind used for published rMD17 results rather than with the fixed protocol of Section S13. For both configurations, we allocate 950 training, 50 validation and 1,000 test configurations using rMD17 predefined index file 01[21], and use conservative forces $F = -\partial E/\partial x$[22], a loss in which the force term carries $100\times$ the weight of the energy term, Adam with weight decay 0.01, at most 1,000 epochs, early stopping with patience 25 on a combined validation metric, and four seeds (42, 43, 44, 45).

- **CliffordSTF-1.9M** (1.87×10^6 parameters, C=80 channels, output MLP width 96) uses the default training configuration released with the GotenNet code[23]: cutoff 5.0 Å, learning rate 10^{-4}, reduce-on-plateau decay 0.8 with patience 5 and floor 10^{-7}, exponential moving average of the weights with decay 0.9, batch size 16.
- **CliffordSTF-10M** (1.017×10^7 parameters) uses its own schedule: cutoff 6.0 Å, learning rate 5×10^{-4} with cosine decay to 10^{-7} over the epoch budget, weight EMA decay 0.999, batch size 8.

The published GotenNet rMD17 experiments use energy-gradient forces with AdamW, warm-up, a peak learning rate of 2×10^{-4}, batch size 4, dropout and up to 3,000 epochs, and average five splits[23]. Our runs match the force formulation but use split 01 and the settings above, including different loss weights. Table S7 therefore provides reference force errors across these protocols.

Comparison with published results. Table S7 places both configurations next to the eight published entries of Table 4 of ref. 23. At 10^7 parameters CliffordSTF reaches a median force MAE of 0.208 kcal mol^{-1} Å^{-1}, $1.23\times$ the highest published entry per molecule (median over molecules) and below it on azobenzene; the leading published models remain several times more accurate.

Per-seed results. Table S8 lists every one of the 80 held-out evaluations with its stopping epoch. All values are computed on the 1,000-configuration test set from the best-validation checkpoint of each run with a single evaluation script that reads each run's own cutoff from its configuration. Mean force cosine lies between 0.9954 and 1.0000 for every evaluation, indicating high directional agreement across runs.

Seed sensitivity and stopping epochs. The mean relative standard deviation of force MAE across four seeds is 31.7% for CliffordSTF-1.9M and 16.2% for CliffordSTF-10M. Across the ten molecules, relative spread correlates with the ratio of longest to

Table S5 OC20 validation results. (a) S2EF energy MAE, force MAE and force cosine similarity; (b) IS2RE relaxed-energy MAE. Mean ± standard deviation; best value per column in bold, second best underlined. Baselines from ref. 19; CliffordSTF averages all 3 seeds. Energy MAE in eV and force MAE in eV Å^{-1}.

(a) S2EF

Model	e_{MAE} ID	e_{MAE} OOD-Ads.	e_{MAE} OOD-Cat.	e_{MAE} OOD-Both	f_{MAE} ID	$f_{\cos}$ ID↑
CliffordSTF (this work)	<u>2.11</u> ± 0.05	<u>2.21</u> ± 0.06	**2.84** ± 0.11	**3.49** ± 0.13	<u>0.060</u> ± 0.000	0.14 ± 0.00
SchNet	4.77 ± 0.05	4.82 ± 0.08	4.85 ± 0.08	5.83 ± 0.08	0.070 ± 0.001	0.13 ± 0.00
DimeNet++	4.10 ± 0.01	4.16 ± 0.00	4.47 ± 0.00	5.37 ± 0.03	0.063 ± 0.000	**0.17** ± 0.00
PaiNN	4.22 ± 0.03	4.29 ± 0.04	4.45 ± 0.01	5.34 ± 0.01	0.066 ± 0.000	0.15 ± 0.00
TorchMD-Net	3.71 ± 0.01	3.76 ± 0.01	4.15 ± 0.01	4.94 ± 0.01	0.064 ± 0.000	<u>0.16</u> ± 0.00
NequIP	5.75 ± 0.08	5.81 ± 0.08	5.90 ± 0.02	7.12 ± 0.01	0.077 ± 0.000	0.11 ± 0.00
GotenNet	3.56 ± 0.20	3.65 ± 0.17	3.96 ± 0.13	4.77 ± 0.15	0.063 ± 0.002	**0.17** ± 0.00
EquiformerV2 L=2	2.47 ± 0.08	2.52 ± 0.09	2.96 ± 0.10	3.62 ± 0.12	**0.059** ± 0.001	**0.17** ± 0.01
EquiformerV2 L=4	2.37 ± 0.12	2.42 ± 0.12	<u>2.89</u> ± 0.10	<u>3.51</u> ± 0.15	<u>0.060</u> ± 0.000	**0.17** ± 0.00
MACE L=1	5.88 ± 0.25	5.93 ± 0.24	6.02 ± 0.15	7.11 ± 0.19	0.077 ± 0.000	0.10 ± 0.00
MACE L=2	6.04 ± 0.33	6.07 ± 0.32	6.10 ± 0.28	7.15 ± 0.30	0.075 ± 0.000	0.11 ± 0.00
ICTP L=1	6.00 ± 0.09	6.09 ± 0.10	5.92 ± 0.09	7.19 ± 0.08	0.078 ± 0.002	0.10 ± 0.00
ICTP L=2	6.64 ± 0.31	6.68 ± 0.30	6.50 ± 0.33	7.85 ± 0.37	0.077 ± 0.000	0.10 ± 0.00
CliffordIP	**1.89** ± 0.01	**1.98** ± 0.01	2.96 ± 0.03	<u>3.51</u> ± 0.00	0.064 ± 0.000	0.12 ± 0.00

(b) IS2RE

Model	ID	OOD-Ads.	OOD-Cat.	OOD-Both
CliffordSTF (this work)	0.70 ± 0.01	0.81 ± 0.02	0.71 ± 0.00	0.75 ± 0.02
SchNet	0.77 ± 0.01	0.86 ± 0.05	0.76 ± 0.01	0.80 ± 0.04
DimeNet++	0.70 ± 0.01	0.84 ± 0.01	<u>0.69</u> ± 0.01	0.77 ± 0.02
PaiNN	0.72 ± 0.01	0.91 ± 0.05	0.71 ± 0.01	0.83 ± 0.04
TorchMD-Net	0.72 ± 0.01	0.81 ± 0.04	0.71 ± 0.02	0.74 ± 0.04
NequIP	0.93 ± 0.03	0.98 ± 0.05	0.90 ± 0.03	0.90 ± 0.05
GotenNet	<u>0.69</u> ± 0.01	**0.77** ± 0.04	**0.67** ± 0.01	**0.71** ± 0.04
EquiformerV2 L=2	**0.65** ± 0.01	<u>0.79</u> ± 0.02	**0.67** ± 0.00	<u>0.72</u> ± 0.02
EquiformerV2 L=4	0.77 ± 0.10	0.85 ± 0.05	0.74 ± 0.10	0.78 ± 0.05
MACE L=1	0.97 ± 0.10	1.05 ± 0.01	0.94 ± 0.09	0.97 ± 0.02
MACE L=2	0.92 ± 0.08	0.95 ± 0.11	0.89 ± 0.07	0.86 ± 0.08
ICTP L=1	0.88 ± 0.01	0.96 ± 0.05	0.85 ± 0.01	0.88 ± 0.04
ICTP L=2	0.88 ± 0.02	1.05 ± 0.09	0.83 ± 0.01	0.93 ± 0.06
CliffordIP	0.77 ± 0.02	0.91 ± 0.03	0.76 ± 0.02	0.85 ± 0.02

Table S6 OC22 validation results. (a) S2EF energy MAE, force MAE and force cosine similarity; (b) IS2RE relaxed-energy MAE. Mean ± standard deviation; best value per column in bold, second best underlined. Baselines from ref. 19; CliffordSTF averages all 4 seeds. Energy MAE in eV and force MAE in eV Å^{-1}. One IS2RE seed diverged early (13.28 eV ID); the other three give 4.09 eV ID.

	(a) S2EF				**(b) IS2RE**	
Model	e_{MAE} ID	e_{MAE} OOD	f_{MAE} ID	$f_{\cos}$ ID↑	ID	OOD
CliffordSTF (this work)	<u>2.92</u> ± 0.11	5.17 ± 0.19	<u>0.073</u> ± 0.000	0.05 ± 0.01	6.39 ± 3.98	9.35 ± 5.00
SchNet	7.35 ± 0.99	11.34 ± 1.47	0.077 ± 0.001	0.03 ± 0.01	5.85 ± 1.25	9.07 ± 0.73
DimeNet++	5.36 ± 0.10	8.85 ± 0.20	0.075 ± 0.000	<u>0.06</u> ± 0.00	5.77 ± 2.51	9.19 ± 2.00
PaiNN	5.41 ± 0.14	8.13 ± 0.97	0.076 ± 0.000	0.05 ± 0.00	<u>3.64</u> ± 0.24	6.85 ± 0.22
TorchMD-Net	4.28 ± 0.00	6.23 ± 0.00	0.075 ± 0.000	0.05 ± 0.00	5.48 ± 0.33	8.10 ± 0.35
NequIP	9.90 ± 0.09	15.88 ± 0.11	0.079 ± 0.000	0.03 ± 0.00	13.95 ± 3.42	16.20 ± 2.84
GotenNet	4.52 ± 0.28	7.21 ± 0.07	<u>0.073</u> ± 0.000	0.05 ± 0.00	4.41 ± 0.21	**6.00** ± 0.44
EquiformerV2 L=2	3.13 ± 0.18	**4.97** ± 0.07	**0.072** ± 0.000	**0.07** ± 0.00	4.63 ± 1.57	6.72 ± 0.93
EquiformerV2 L=4	3.38 ± 0.04	5.26 ± 0.04	**0.072** ± 0.000	**0.07** ± 0.00	5.30 ± 0.64	6.78 ± 0.34
MACE L=1	10.11 ± 0.14	16.00 ± 0.45	0.078 ± 0.000	0.02 ± 0.00	10.87 ± 3.35	14.07 ± 2.82
MACE L=2	10.26 ± 0.41	15.82 ± 0.74	0.078 ± 0.000	0.02 ± 0.00	22.65 ± 20.13	38.52 ± 22.99
ICTP L=1	11.70 ± 0.59	17.52 ± 0.53	0.080 ± 0.001	0.02 ± 0.00	10.11 ± 1.53	20.77 ± 0.50
ICTP L=2	12.37 ± 0.23	18.72 ± 0.72	0.081 ± 0.001	0.02 ± 0.00	9.91 ± 2.05	17.12 ± 2.56
CliffordIP	**2.89** ± 0.21	<u>5.12</u> ± 0.23	<u>0.073</u> ± 0.000	0.04 ± 0.00	**3.47** ± 0.25	<u>6.58</u> ± 0.41

Table S7 rMD17 force MAE under a published training recipe (kcal mol^{-1} Å^{-1}, test set, mean ± standard deviation over four seeds). CliffordSTF-1.9M uses the released GotenNet rMD17 configuration; CliffordSTF-10M uses its own schedule (Section S15). Both use split 01 and a force-weighted loss, whereas the published entries average five splits. Published values are taken from Table 4 of ref. 23; "best" and "widest" are the lowest and highest of the eight published entries for each molecule. Force cosine similarity lies between 0.9954 and 1.0000 for all 80 CliffordSTF evaluations.

Molecule	CliffordSTF-10M	CliffordSTF-1.9M	Best published	Widest published	GotenNet	10M/widest	10M/GotenNet
Aspirin	0.515 ± 0.015	1.149 ± 0.241	0.1306 (GotenNet)	0.4128 (ACE)	0.1306	1.25	3.95
Azobenzene	0.196 ± 0.045	0.903 ± 0.302	0.0483 (GotenNet)	0.2514 (ACE)	0.0483	**0.78**	4.05
Benzene	0.091 ± 0.068	0.269 ± 0.136	0.0046 (Allegro)	0.0168 (UNiTE)	0.0047	5.40	19.30
Ethanol	0.221 ± 0.009	0.589 ± 0.228	0.0482 (GotenNet)	0.1683 (ACE)	0.0482	1.31	4.58
Malonaldehyde	0.354 ± 0.042	0.711 ± 0.177	0.0830 (Allegro)	0.2560 (ACE)	0.0830	1.38	4.27
Naphthalene	0.122 ± 0.019	0.753 ± 0.352	0.0208 (Allegro)	0.1176 (ACE)	0.0240	1.04	5.10
Paracetamol	0.370 ± 0.009	0.758 ± 0.152	0.0928 (GotenNet)	0.2929 (ACE)	0.0928	1.26	3.99
Salicylic acid	0.254 ± 0.014	0.757 ± 0.262	0.0669 (Allegro)	0.2145 (ACE)	0.0703	1.19	3.62
Toluene	0.156 ± 0.014	0.669 ± 0.145	0.0261 (GotenNet)	0.1499 (ACE)	0.0261	1.04	5.98
Uracil	0.185 ± 0.023	0.573 ± 0.144	0.0415 (Allegro)	0.1522 (ACE)	0.0417	1.21	4.43
Median	0.208	0.732				1.23	4.35

Table S8 Per-seed rMD17 test force MAE (kcal mol^{-1} Å^{-1}) for both CliffordSTF configurations under the published-recipe protocol, with the stopping epoch of each run and the minimum force cosine similarity across the four seeds. "Ratio" is the longest divided by the shortest stopping epoch; "rel. std" is the standard deviation divided by the mean.

Config.	Molecule	Seed 42	Seed 43	Seed 44	Seed 45	Stopping epochs	Ratio	Rel. std / min cos
10M	Aspirin	0.4997	0.5066	0.5206	0.5342	459, 482, 497, 487	1.08	3.0% / 0.9994
	Azobenzene	0.1712	0.1742	0.2629	0.1747	644, 946, 156, 531	6.06	22.9% / 0.9998
	Benzene	0.0352	0.0304	0.1650	0.1322	364, 590, 64, 133	9.22	75.2% / 1.0000
	Ethanol	0.2242	0.2086	0.2212	0.2293	487, 663, 371, 472	1.79	4.0% / 0.9998
	Malonaldehyde	0.3284	0.4156	0.3436	0.3291	632, 160, 423, 586	3.95	11.7% / 0.9996
	Naphthalene	0.1234	0.1003	0.1201	0.1455	303, 1000, 649, 195	5.13	15.1% / 1.0000
	Paracetamol	0.3650	0.3803	0.3748	0.3613	585, 348, 293, 450	2.00	2.4% / 0.9996
	Salicylic acid	0.2563	0.2596	0.2670	0.2346	621, 1000, 750, 980	1.61	5.5% / 0.9998
	Toluene	0.1772	0.1502	0.1452	0.1516	216, 438, 369, 453	2.10	9.2% / 0.9999
	Uracil	0.1696	0.1619	0.2132	0.1940	422, 697, 418, 489	1.67	12.7% / 0.9999
1.9M	Aspirin	1.1489	0.8221	1.3942	1.2313	324, 1000, 189, 218	5.29	20.9% / 0.9954
	Azobenzene	1.1837	0.5972	1.1407	0.6917	365, 227, 138, 178	2.64	33.4% / 0.9970
	Benzene	0.4706	0.2271	0.1955	0.1827	243, 227, 226, 252	1.12	50.5% / 0.9996
	Ethanol	0.6225	0.3257	0.8750	0.5319	337, 1000, 330, 308	3.25	38.7% / 0.9981
	Malonaldehyde	0.8997	0.5353	0.5864	0.8210	255, 862, 999, 219	4.56	24.9% / 0.9986
	Naphthalene	0.8463	0.4970	0.4567	1.2114	311, 199, 633, 86	7.36	46.8% / 0.9988
	Paracetamol	0.7975	0.7922	0.5415	0.8998	386, 278, 1000, 275	3.64	20.1% / 0.9986
	Salicylic acid	1.0747	0.6769	0.4500	0.8248	217, 342, 1000, 284	4.61	34.7% / 0.9977
	Toluene	0.8716	0.5263	0.6466	0.6309	180, 275, 373, 229	2.07	21.7% / 0.9980
	Uracil	0.6491	0.5090	0.4050	0.7292	256, 282, 1000, 241	4.15	25.2% / 0.9991

shortest stopping epoch for CliffordSTF-10M (Pearson r=0.90), but not for CliffordSTF-1.9M (r=−0.03; Fig. S1). The four 10M benzene runs stop at epochs 64, 133, 364 and 590, with force MAE decreasing from 0.165 to 0.030 kcal mol^{-1} Å^{-1} in that order. In contrast, the 1.9M benzene runs stop within epochs 226–252 yet have a 50.5% relative spread. These observations associate stopping behaviour with error variability in the larger configuration; determining the effect of a longer patience would require a controlled comparison. Mean force cosine remains above 0.995 for every evaluation.

Memory. Peak GPU memory on rMD17 aspirin (batch 32, 672 atoms and 11,456 edges per batch, RTX 4090, PyTorch 2.4.1) is 408 MiB at inference and 1,962 MiB in training for CliffordIP (1.39×10^6 parameters), and 491 MiB and 2,596 MiB for CliffordSTF (1.06×10^6 parameters): the dual-track scaffold costs 20% more memory at inference and 32% more in training at a slightly smaller parameter count.

S16 Wall-clock measurements at matched parameter budget

Table S9 reports rMD17 aspirin timings at approximately 10^6 parameters, batch size 32, on a single GPU. Measurements use 10 warm-up steps followed by 20 timed steps. CliffordSTF takes 0.352 s per inference step and 7.77 s per training step, versus 0.128 and 3.11 s for CliffordIP. Its inference time is comparable to ICTP L=2/3; training takes 1.25× as long as ICTP L=3 and 3.13× as long as EquiformerV2 L=4. The forward/backward decomposition is 2.59/5.10 s for CliffordSTF and 1.09/1.32 s for EquiformerV2 L=4, indicating a larger backward-pass cost in the dual-track implementation. These timings describe the measured implementations and hardware.

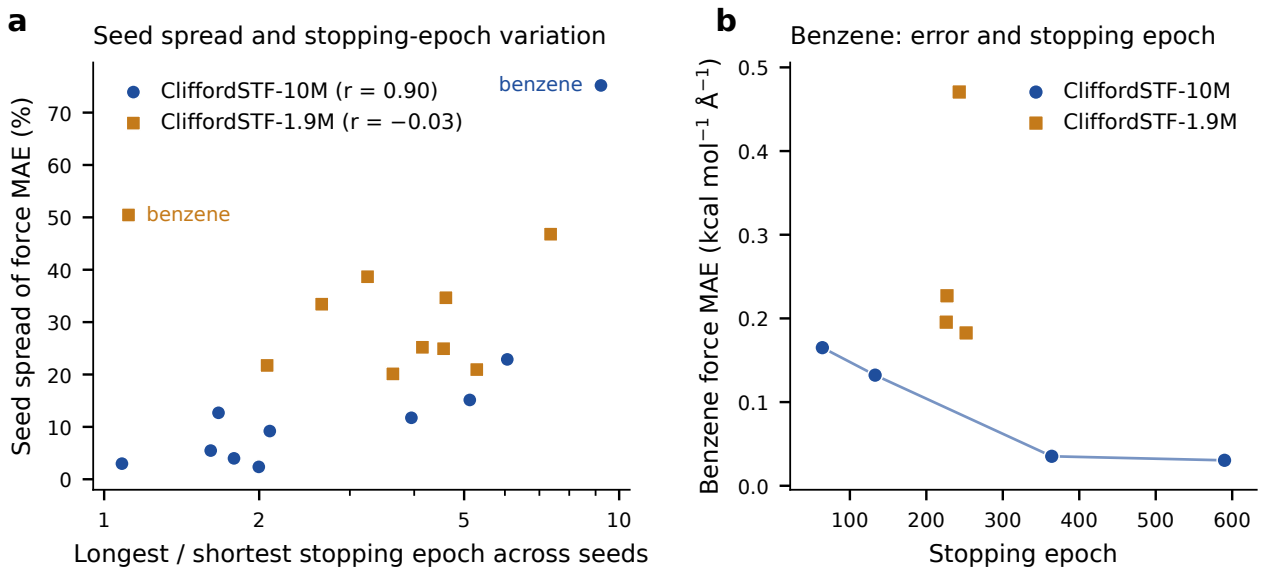


Fig. S1 Seed spread and early stopping on rMD17. (a) Relative standard deviation of test force MAE across four seeds against the ratio of the longest to shortest stopping epoch, one point per molecule (n=10 per configuration); r is the Pearson correlation. (b) Benzene test force MAE against stopping epoch for each of the four seeds; the line joins the CliffordSTF-10M seeds in order of run length. Mean force cosine similarity exceeds 0.995 for all 80 evaluations.

Table S9 Wall-clock at matched parameter budget, rMD17 aspirin, batch 32, single GPU. Time columns are mean ± standard deviation over 20 timed steps following 10 warmup steps; samples-per-second columns are derived from the means. Lower is faster for the time columns; higher is faster for the samples-per-second columns. All models fall within ±50% of the 10^6-parameter target. The Clifford and CliffordSTF (ours) entries reflect the optimized PyTorch dispatch described in Section S17.

Model	Params	Train step (s)	Train (s/s)	Infer step (s)	Infer (s/s)
PaiNN	1,043,856	0.79 ± 0.05	1,145	0.028 ± 0.003	3,575
SchNet	984,769	1.16 ± 0.02	773	0.069 ± 0.011	1,456
FAENet	1,027,745	1.69 ± 0.03	533	0.111 ± 0.004	900
GotenNet	1,057,729	1.81 ± 0.00	497	0.064 ± 0.005	1,571
MACE L=1	1,006,014	1.81 ± 0.29	497	0.160 ± 0.008	624
EquiformerV2 L=2	891,560	2.45 ± 0.01	367	0.131 ± 0.001	764
EquiformerV2 L=3	1,041,304	2.45 ± 0.01	367	0.135 ± 0.003	741
EquiformerV2 L=4	1,043,222	2.48 ± 0.00	363	0.142 ± 0.008	702
MACE L=2	1,011,683	2.53 ± 0.09	355	0.237 ± 0.002	422
TorchMD-NET	968,242	2.58 ± 0.02	349	0.186 ± 0.029	538
NequIP	1,002,436	3.06 ± 0.04	295	0.221 ± 0.007	452
CliffordIP	1,031,926	3.11 ± 0.01	289	0.128 ± 0.000	781
ViSNet	957,606	3.21 ± 0.07	280	0.235 ± 0.001	425
MACE L=3	980,092	3.60 ± 0.16	250	0.261 ± 0.009	384
ICTP L=1	1,039,554	4.50 ± 0.11	200	0.302 ± 0.052	331
ICTP L=2	997,368	5.38 ± 0.10	167	0.364 ± 0.007	275
DimeNet++	1,064,070	5.84 ± 0.11	154	0.326 ± 0.002	307
ICTP L=3	982,752	6.23 ± 0.02	144	0.356 ± 0.006	281
CliffordSTF (ours)	1,056,534	7.77 ± 0.11	116	0.352 ± 0.001	284

S17 Implementation optimizations

The timings use standard PyTorch operations[24] without custom CUDA or Triton kernels. The geometric product is evaluated by an `einsum` against a precomputed Cayley structure tensor. Graph counts are passed explicitly to avoid per-layer host synchronisation, layer-independent routing descriptors are computed once per forward pass, and validation metrics accumulate on device. Compilation is disabled in the reported configuration.

S18 Per-seed stability evidence for cross-track coupling

We compare the existing *Full* and *Full no cross* runs using per-run errors, per-seed means and a count of unstable runs. The stored configurations show that *Full no cross* disables both cross-track coupling and adaptive routing, so this comparison does not isolate coupling alone. Each configuration comprises 24 runs: 4 molecules (aspirin, benzene, ethanol, salicylic acid) × 6 seeds. The metric is

test energy MAE in meV. The mean ± std reported in Table 2 averages the four molecules within each seed and then takes mean and standard deviation over the six seeds; both levels of aggregation are given below.

The *loss ratio* column normalises each run's final training loss by the median final training loss for that molecule across the twelve variants in the original `MD17_Ablation` sheet, including the additional no-routing run series and excluding the later STF-only readout series, giving a molecule-independent scale on which a diverged optimisation is identifiable without reference to the test metric.

Table S10 Per-run energy MAE and normalised final training loss, Full versus Full no cross on rMD17. Loss ratio is the run's final training loss divided by the median final training loss for that molecule across the twelve original `MD17_Ablation` variants. The single run with loss ratio above 5 (aspirin, seed 4, no cross) is the only diverged optimisation in either configuration.

		Full (cross on)		**Full, no cross**	
Molecule	Seed	e_{MAE} (meV)	Loss ratio	e_{MAE} (meV)	Loss ratio
Aspirin	1	51.74	0.52	24.08	0.45
Aspirin	2	11.55	1.11	42.16	0.71
Aspirin	3	69.21	0.66	57.93	2.80
Aspirin	4	7.93	1.11	963.07	25.24
Aspirin	5	12.01	3.09	24.42	1.78
Aspirin	6	72.88	2.45	17.21	0.88
Benzene	1	25.72	0.87	12.08	1.27
Benzene	2	24.35	2.05	61.82	2.31
Benzene	3	8.85	1.47	30.29	1.37
Benzene	4	30.75	0.77	32.89	1.45
Benzene	5	550.56	2.77	42.94	1.26
Benzene	6	6.96	0.47	46.87	2.28
Ethanol	1	3.04	0.86	7.58	0.93
Ethanol	2	32.70	4.71	16.70	0.92
Ethanol	3	2.86	0.54	14.22	1.09
Ethanol	4	14.00	0.83	11.96	0.60
Ethanol	5	12.31	0.54	33.78	1.32
Ethanol	6	12.52	0.87	13.60	0.99
Salicylic	1	29.71	0.56	74.36	1.27
Salicylic	2	38.07	0.51	56.54	1.10
Salicylic	3	54.69	0.81	29.86	0.71
Salicylic	4	21.60	0.70	23.34	1.01
Salicylic	5	4.29	1.09	10.11	1.18
Salicylic	6	5.36	0.95	42.01	1.62

Table S11 Seed-level energy MAE (molecule-averaged, the aggregation used in Table 2) and summary statistics at both aggregation levels.

	Full (cross on)	**Full, no cross**
Seed-level e_{MAE} (meV), molecules averaged within seed		
1	27.55	29.52
2	26.67	44.30
3	33.90	33.08
4	18.57	257.81
5	144.79	27.81
6	24.43	29.92
Seed-level median	27.11	31.50
Seed-level mean ± std	45.99 ± 48.66	70.41 ± 92.00
Run-level statistics (all 24 runs)		
Median e_{MAE} (meV)	17.80	30.07
Interquartile range (meV)	8.6–34.0	16.1–43.9
Maximum e_{MAE} (meV)	550.56	963.07
Runs with $e_{\mathrm{MAE}} > 50$ meV	5/24	5/24
Runs with $e_{\mathrm{MAE}} > 100$ meV	1/24	1/24
Runs with loss ratio > 5	**0/24**	**1/24**
Mean ± std excluding worst run (meV)	24.05 ± 20.86	31.60 ± 18.69
Median $f_{\cos}$	0.547	0.490

Summary. In the full STF_2+STF_3 configuration, the observed divergence count is 0/24 runs with coupling and 1/24 without it, unchanged at loss-ratio thresholds of 5, 10 and 20. Median run-level energy MAE is 17.8 versus 30.1 meV, with coupling lower in 16 of 24 matched molecule–seed pairs. The paired differences are not statistically significant at this sample size (Wilcoxon signed-rank test[25]: $p = 0.229$ for energy MAE and $p = 0.623$ for force cosine). The two configurations each contain one large test-energy outlier; the coupled benzene outlier has a loss ratio of 2.77 and is not classified as a training divergence. Divergence is therefore defined from training loss rather than test energy error.

The result is configuration-dependent. Removing coupling from the STF_2-only model improves mean energy MAE from 34.9 to 25.8 meV and force cosine from 0.421 to 0.456.

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